STUDY GUIDE · TEAM SOLOX

Signal Grid Mapping

Split a space into cells, connect them with links, collect small signals, and let the pattern show what is there, where it moves, and what is missing. Every resource you need to learn it, in the web stack and beyond.

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Aman Kumar Chhari · October 2026 · Links are clickable in the PDF.

FREE free to use or read PAID book or course with a cost START best first step

1 The pattern in one page

You do not search for the hidden thing. You wait for the signals it leaves, collected in a fixed structure, and the thing shows itself.

Cells
The area split into equal hexagons. Each cell senses what happens inside it.

Links
Lines between cells along paths that matter. They sense what moves, and which way.

Signals
Small reports: where, when, what kind, how strong.

Hub
One place that receives every signal, combines them and shows the picture.

Divide space into cells → **Connect** cells with links → **Collect** signals with time → **Combine** score, fade → **Decide** threshold, confirm → **Show** glow, arrows → **Act** send help or fix

WHAT YOU SEE	MEANING
Dark cell	Nothing happening
Glowing cell	Something is present; brighter means more activity
Glow moving cell to cell	Something is moving, and you can see which way
Links lighting in a row	A clear path, so the next cell can be predicted
Glow that stops	It closed, ended or left
Many requests, no glow	A gap: something is needed here and missing

Six rules

1. **Count signals, not people.** Show totals, never identities.
2. **One signal is noise; repeated signals are a pattern.**
3. **Two different sources beat one.** A payment plus a delivery is stronger than either alone.
4. **Everything fades.** Old signals lose strength (half-life), so the map shows now.
5. **Place links where things must pass.** Choke points give the clearest signals.
6. **Show what is missing.** A gap is as useful as a glow.

The core maths (all you need to start)

```
cell     = h3.latLngToCell(lat, lng, 9)                // ~city-block hexagon
score    = score × 0.5^(secondsSinceLast / halfLife) + weight   // fade, then add
active   = score ≥ threshold  AND  distinctKinds ≥ 2      // confirmed
gap      = demandScore ≥ gapThreshold AND score < 1       // wanted, nothing here
link     = track moved cell A → cell B  ⇒  flow(A→B) += 1 // movement
nextCell = link out of current cell with the highest flow  // prediction
```

Each topic in this guide maps to one of these lines: hexagon grids → *cell*; event streams → *collect*; spatial statistics → *active* and *gap*; tracking → *link* and *nextCell*.

2 Study path (8 weeks)

About 5–7 hours a week. Each week ends with something you can show.

WEEK	LEARN	BUILD
1 · Grids	Red Blob Games hexagons; H3 concepts (resolutions, rings, parents)	A page that draws H3 hexagons around a fixed point
2 · Maps	MapLibre GL JS basics; deck.gl layers; map styles and tiles	MapLibre + deck.gl `H3HexagonLayer` with your GPS location (`navigator.geolocation`)
3 · Geo maths	Turf.js: distance, circles, point-in-polygon, clustering	Ripple search: 500 m → 1 km → 2 km until 3 places are found
4 · Signals and events	Event sourcing, CloudEvents, SSE vs WebSockets (Kleppmann ch. 11)	Next.js API: `POST /signals`, `GET /stream` (SSE) updating the map live
5 · Storage	PostGIS basics, h3-pg, Redis GEO `GEOSEARCH`	Supabase table of signals with fade, threshold and gap queries
6 · Hotspots	Kernel density, Getis-Ord Gi*, Moran's I, spatial scan statistic	Compare your simple threshold with a Gi* hotspot map in QGIS or PySAL
7 · Movement	Occupancy grids (Probabilistic Robotics ch. 9), Kalman filter basics	deck.gl `ArcLayer` and `TripsLayer`, next-cell prediction
8 · Real world	Read about Waze, DART buoys, HealthMap; location privacy	Simulator with hidden "truth", measure hit rate and false alarms, write it up

> **Tip:** always test with a simulator first: hide a few "true" places, generate noisy signals, and check whether the grid finds them. Hit rate and false alarms are your proof that the method works.

3 Start here

Read these first. Together they take an afternoon and give you the whole mental model.

RESOURCE	WHAT YOU LEARN
Red Blob Games: Hexagonal Grids START FREE	Hexagon coordinates, neighbours, distance, rings, pixel ↔ hex. Interactive, with JS examples. The best introduction anywhere.
H3 documentation FREE	Uber's global hexagon index: 16 resolutions, `gridDisk` rings, parent/child cells, cell areas table.
Uber blog: H3, Uber's Hexagonal Hierarchical Spatial Index FREE	Why hexagons beat squares for supply, demand and pricing.
John Snow's 1854 cholera map FREE	The original "hidden cause from scattered signals" story.
Occupancy grid mapping (overview) FREE	The robotics version of the same idea: grid cells that collect evidence about what is there.
NOAA DART tsunami buoys FREE	A real network of ocean sensors detecting hidden waves: the buoy idea in practice.

4 Books

BOOK	WHY READ IT
Geocomputation with Python (Dorman, Graser, Nowosad, Lovelace) FREE	Practical spatial data: points, polygons, rasters, maps. Good even if you mostly use JS, for the concepts.
Geocomputation with R (Lovelace, Nowosad, Muenchow) FREE	The original, more complete edition of the same book.
Spatial Data Science (Pebesma & Bivand) FREE	Theory of spatial data, coordinate systems, point patterns, spatial statistics.
Geographic Information Science and Systems (Longley, Goodchild, Maguire, Rhind) PAID	The standard GIS textbook: how space is represented and analysed.
Probabilistic Robotics (Thrun, Burgard, Fox) PAID	Chapter 9: occupancy grid mapping. Also Kalman and particle filters for tracking. The scientific heart of the pattern.
Wireless Sensor Networks: An Information Processing Approach (Zhao & Guibas) PAID	Detecting and tracking targets with many small sensors: cells and links at hardware level.
Designing Data-Intensive Applications (Martin Kleppmann) PAID	Chapter 11 (stream processing): event logs, turning events into state. This is your "hub".
GeoDa Center workbook and docs (Luc Anselin) FREE	Hands-on spatial autocorrelation, hotspots and clusters with a free desktop tool.

5 Research and methods

Search the title in Google Scholar. Each one answers one question your grid has to answer.

QUESTION	METHOD / PAPER
How do cells collect evidence?	Alberto Elfes, "Using Occupancy Grids for Mobile Robot Perception and Navigation", *IEEE Computer*, 1989. Moravec & Elfes, "High Resolution Maps from Wide Angle Sonar", ICRA 1985.
Where are the hotspots?	Getis & Ord, "The Analysis of Spatial Association by Use of Distance Statistics", 1992 (Gi*). Kernel density estimation.
Is a cluster real or chance?	Martin Kulldorff, "A Spatial Scan Statistic", 1997. Free tool: SaTScan.
Do nearby cells behave alike?	Moran's I; Anselin, "Local Indicators of Spatial Association: LISA", 1995.
How do I track something moving?	Kalman filter (Kalman, 1960); particle filters; multi-target tracking. Easy start: kalmanfilter.net FREE.
Can behaviour reveal events?	Ginsberg et al., "Detecting influenza epidemics using search engine query data", *Nature*, 2009, and the warning that followed: Lazer et al., "The Parable of Google Flu", *Science*, 2014.
How do agents ask and offer?	Reid G. Smith, "The Contract Net Protocol", *IEEE Trans. Computers*, 1980.
How do I protect location privacy?	Andrés et al., "Geo-Indistinguishability: Differential Privacy for Location-Based Systems", ACM CCS 2013.
How is the world indexed in cells?	Sahr, White & Kimerling, "Geodesic Discrete Global Grid Systems", 2003. Google's S2 cells as the square alternative to H3.

6 Courses

COURSE / CHANNEL	COVERS
Coursera: *Geographic Information Systems (GIS) Specialization*, UC Davis PAID **(audit free)**	GIS foundations, spatial analysis, real projects.
Coursera: *GIS, Mapping, and Spatial Analysis Specialization*, University of Toronto PAID **(audit free)**	Mapping, spatial analysis, remote sensing.
YouTube: Luc Anselin, GeoDa lectures FREE	Spatial autocorrelation, hotspots, clusters: the statistics behind "glow".
YouTube: Cyrill Stachniss, robotics / SLAM lectures FREE	Occupancy grids, Bayes filters, Kalman and particle filters.
YouTube: Steve Brunton (Control Bootcamp) FREE	Kalman filter intuition and estimation.
YouTube: 3Blue1Brown FREE	Bayes' theorem and probability intuition for combining signals.
YouTube: Uber Engineering / vis.gl talks on deck.gl and H3 FREE	How large live maps are built.
QGIS Tutorials and Tips (Ujaval Gandhi) FREE	Step-by-step heatmaps, hex grids and hotspot maps without code.
Spatial Thoughts courses FREE	Free GIS and Python geospatial courses by the same author, India-based examples.

7 YouTube

All free. Found on YouTube in October 2026; titles as listed there. Watch the START video of each group first.

The idea: hexagon grids and H3

VIDEO	WHY WATCH
H3: Tiling the Earth with Hexagons (Uber Engineering) START	Why Uber indexes the world in hexagons to balance riders and drivers.
What is a Spatial Index? (H3 Explained)	Short explainer of spatial indexes and H3 cells.
How Does Uber Find Nearby Drivers	The "who is near me" problem, the heart of ripple search.
Engineering Sub-City Geos for a Hyper-Local Marketplace	Isaac Brodsky and Nick Rabinowitz on H3 at Uber (2019 meetup).
Engineering an H3-based Geospatial Data Platform at Uber	How a whole data platform is built on H3 cells.

Building it on the web

VIDEO	WHY WATCH
H3-js: Hexagons in the Browser (Uber Open Summit 2018) START	Using h3-js in web apps: your exact stack.
Geospatial Analytics with H3 + kepler.gl (Isaac Brodsky)	Hexagon analysis and visualisation together.
Large-scale Geospatial Visualization with kepler.gl (Shan He)	How kepler.gl and deck.gl render huge map data.
Data visualization with deck.gl, part 1	deck.gl layers from scratch (with the Google Maps JS API).
Data visualization with deck.gl, part 2	More layers and interaction.
MapLibre + deck.gl Deeper Integration	How the two libraries work together.
Migrating visualizations from MapLibre to deck.gl (Global Fishing Watch)	A real live map of moving things, built on this stack.
Server-Sent Events Crash Course	SSE vs polling vs WebSockets: the live stream to your map.

What are Server-Sent Events (SSE)? (Ably)	Short SSE overview.
PostGIS Introduction (Paul Ramsey, PostGIS Day 2019)	PostGIS from one of its core developers.
Spatial SQL, the Modern Data Stack, and PostGIS	Where spatial SQL fits in a modern backend.

Finding real hotspots (statistics)

VIDEO	WHY WATCH
Hot Spot Analysis: the Fundamentals (Gi & Gi*) START	How Gi* decides a cluster is real, not chance.
QGIS Spatial Statistics for Beginners: Getis-Ord Gi & Local Moran's I	Hotspots and clusters hands-on, no code.

Grids that collect evidence, and tracking movement

VIDEO	WHY WATCH
Occupancy Grid: 5 Minutes with Cyrill START	The grid-of-evidence idea in five minutes.
Occupancy Grid Maps (Cyrill Stachniss, full lecture)	Full lecture: how each cell updates its belief from signals.
SLAM Course 11: Grid Maps (Cyrill Stachniss)	Deeper version from his SLAM course.
Visually Explained: Kalman Filters	Intuition for tracking a moving thing from noisy signals.
Why Use Kalman Filters? (MathWorks)	Fusing GPS and other sensors to estimate position.
Kalman Filter for Beginners (Shane Ross, playlist)	From moving averages up to the Kalman filter, with code.

The story behind the pattern

VIDEO	WHY WATCH
The Ghost Map	John Snow and the 1854 cholera map.
Ghosts, Germs and Maps: The Legacy of the 1854 London Cholera Epidemic	How one map started modern epidemiology.

Channels worth following

CHANNEL	FOR
Cyrill Stachniss	Occupancy grids, SLAM, Bayes and Kalman filters, "5 Minutes with Cyrill" short explainers.
Steve Brunton	Control Bootcamp: Kalman filters and estimation.
3Blue1Brown	Bayes' theorem and probability intuition for combining signals.
Luc Anselin (GeoDa)	Spatial statistics lectures: autocorrelation, LISA, hotspots.
Uber Engineering / Uber Open Source	H3, kepler.gl and deck.gl talks.

> **Watch order (about 4 hours):** H3: Tiling the Earth → H3-js in the Browser → Occupancy Grid in 5 minutes → Hot Spot Fundamentals → SSE Crash Course → deck.gl parts 1–2 → Visually Explained: Kalman Filters → The Ghost Map.

8 Real systems to study

Each uses the same pattern. Read one engineering post or paper about each.

SYSTEM	THE PATTERN INSIDE IT
NOAA DART buoys	Ocean-floor sensors detect pressure changes and reveal tsunamis.
USGS earthquake network (earthquake.usgs.gov)	Many sensors locate a hidden event from arrival times.
Waze	Phones slowing down reveal jams and accidents on road links.
Uber / Ola surge pricing	H3 cells of supply and demand; gaps raise prices.
HealthMap, BlueDot	Outbreaks spotted early from scattered reports and news.
Global Forest Watch	Grid cells light up when satellites see forest loss.
Flightradar24	A live map built from thousands of small receivers.
Google Maps "Popular times"	Anonymous phone signals show how busy a place is.
ONDC / Beckn (becknprotocol.io)	Open search → offer → confirm network in India: the connection protocol.

9 Web stack: tools

Everything in TypeScript. Bold items are the recommended default.

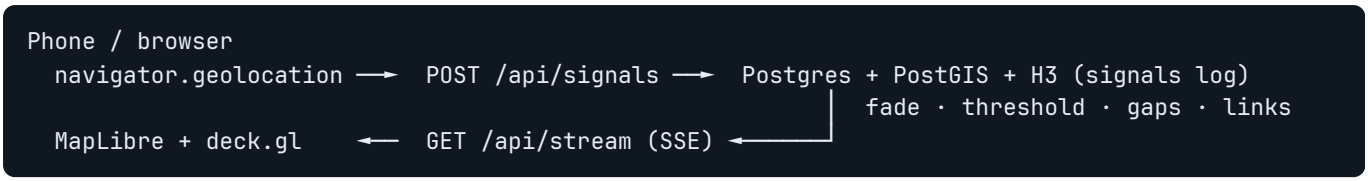
Frontend

JOB	TOOL
Hexagon cells	**h3-js**; or honeycomb-grid for local, non-geographic grids
Distance, circles, inside-polygon, clusters	**Turf.js** (`distance`, `circle`, `hexGrid`, `pointsWithinPolygon`, `clustersDbscan`)
Base map	**MapLibre GL JS** (free vector maps) or Leaflet (simple)
Grid, arcs, heatmaps, trails	**deck.gl**: `H3HexagonLayer`, `ArcLayer`, `TripsLayer`, `HeatmapLayer`
React bindings	react-map-gl + `@deck.gl/react`
Real Google Maps	Google Maps JavaScript API + `@deck.gl/google-maps` (needs an API key; the allowed way)
Free map styles	OpenFreeMap, Protomaps (PMTiles), MapTiler / Stadia free tiers
User location	**navigator.geolocation** (GPS on phones: accurate)
Density and statistics	d3-contour (density), simple-statistics
Tracking	kalmanjs
Heavy maths off the main thread	Web Workers (Comlink makes them easy)

Backend

JOB	TOOL
API	**Next.js route handlers**, Hono, Fastify or Express
Live updates	**SSE** (`text/event-stream`, auto-reconnect with `Last-Event-ID`); Socket.IO when you need two-way
"Who is near me", fast	**Redis GEO** GEOSEARCH (ioredis, Upstash)
Signals and history	**Postgres + PostGIS** + h3-pg, via Drizzle or Prisma
All-in-one	**Supabase**: Postgres, PostGIS, Realtime, Auth
Event format and bus	CloudEvents JS SDK; later NATS.js or Redis Streams

10 Web stack: architecture and code



Signal in, cell out (Next.js route)

```
// app/api/signals/route.ts
import { latLngToCell } from "h3-js";
export async function POST(req: Request) {
  const { kind, lat, lng, track } = await req.json();
  const cell = latLngToCell(lat, lng, 9);
  await db.insert(signals).values({ kind, cell, lat, lng, track, at: new Date() });
  bus.publish({ type: "grid.signal", cell, kind });      // wakes SSE subscribers
  return Response.json({ cell });
}
```

Cell scores with fade (SQL)

```
SELECT cell,
       SUM(weight * POWER(0.5, EXTRACT(EPOCH FROM now() - at) / 1200)) AS score,
       COUNT(DISTINCT kind)                                            AS kinds
FROM signals
WHERE at > now() - interval '3 hours'
GROUP BY cell
HAVING SUM(weight * POWER(0.5, EXTRACT(EPOCH FROM now() - at) / 1200)) > 0.3;
```

Live stream (SSE)

```
// app/api/stream/route.ts
export async function GET() {
  const stream = new ReadableStream({
    start(c) {
      const send = async () => c.enqueue(`data: ${JSON.stringify(await cellStates())}\n\n`);
      bus.subscribe(send); send();
    },
  });
  return new Response(stream, { headers: { "Content-Type": "text/event-stream" } });
}
```

Draw it (deck.gl)

```
import { H3HexagonLayer } from "@deck.gl/geo-layers";
import { ArcLayer } from "@deck.gl/layers";
const cells = new H3HexagonLayer({
  id: "cells", data: states, getHexagon: d ⇒ d.cell,
  getFillColor: d ⇒ d.gap ? [194, 38, 62, 150] : d.active ? [21, 128, 61, 180] : [11, 107,
203, 60],
});
const links = new ArcLayer({
  id: "links", data: flows, getSourcePosition: d ⇒ cellToLatLng(d.from).reverse(),
  getTargetPosition: d ⇒ cellToLatLng(d.to).reverse(), getWidth: d ⇒ 1 + d.flow,
});
new EventSource("/api/stream").onmessage = e ⇒ setStates(JSON.parse(e.data));
navigator.geolocation.watchPosition(p ⇒ send("gps", p.coords.latitude, p.coords.longitude));
```

11 Web learning resources

RESOURCE	FOR
deck.gl examples gallery START FREE	H3 hexagons, arcs, trips, heatmaps, each with source code.
MapLibre GL JS examples FREE	Maps, layers, markers, live GeoJSON updates.
kepler.gl FREE	Drop in a CSV of points and see hexagon maps instantly; great for exploring before coding.
vis.gl FREE	The family behind deck.gl, react-map-gl, loaders.gl.
Turf.js docs FREE	Every geo calculation in JS, with live examples.
Observable notebooks (search "H3", "deck.gl", "d3 density") FREE	Interactive JS examples you can fork and change.
Supabase PostGIS guide FREE	Location queries ("nearby") and realtime in a hosted Postgres.
PostGIS workshop FREE	Spatial SQL from zero: geometry, indexes, `ST_DWithin`.
Redis GEOSEARCH docs FREE	Millisecond "who is within 500 m".
MDN: Server-sent events FREE	The live stream to the map.
MDN: Geolocation API FREE	Accurate user location, permissions, `watchPosition`.
OpenStreetMap + Overpass Turbo FREE	Free real data (shops, roads, places) to practise on.

12 Python and desktop tools

Not needed for the web build, but useful for analysis, statistics and checking your results.

TOOL	USE
QGIS FREE	Desktop GIS: hex grids, heatmaps, hotspots with no code.
GeoDa FREE	Moran's I, LISA, hotspot maps, point and click.
GeoPandas FREE	Tables of places with geometry in Python.
h3-py FREE	H3 in Python (same API as h3-js).
PySAL (esda) FREE	Gi*, Moran's I, LISA in code.
SaTScan FREE	Spatial and space-time cluster detection.
OSMnx FREE	Download real street networks: ready-made "links" between places.

13 Practice projects and checklist

Five projects, easy to hard

1. **My hex neighbourhood:** MapLibre + deck.gl, your GPS location, H3 rings of 1, 2 and 3 around you.
2. **Hidden shops:** simulator hides 5 places, sends noisy signals; your grid must find them. Show hit rate and false alarms.
3. **Live demand gaps:** users tap "I need X here"; red cells where nothing supplies it. Supabase + SSE.
4. **Moving things:** two phones share anonymous GPS; arcs show flow, the next cell is predicted.
5. **Second domain, no code change:** swap the settings (cell size, signals, threshold) for water leaks or traffic.

Checklist: I can...

☐ explain why hexagons beat squares (equal neighbours, equal distances)
☐ convert lat/lng to an H3 cell and get its ring of neighbours
☐ draw H3 cells on MapLibre with deck.gl
☐ get accurate user location with `navigator.geolocation`
☐ store signals in PostGIS and query cells with fading scores
☐ push live updates over SSE and resume after a reconnect
☐ tell a real hotspot from noise (threshold + two sources, or Gi*)
☐ show movement with links and predict the next cell
☐ find gaps: demand without supply
☐ protect privacy: counts per cell, never identities
☐ measure the grid against a hidden truth (hit rate, false alarms)
☐ add a second domain with only a settings change

> **One line to remember:** whatever you cannot see still leaves signals. Divide the world into cells, connect them with links, let each report what it feels, and the pattern shows what is there, where it is going, and what is missing.